



RESEARCH COMIC

A FAIR Comic about Research Data Infrastructure

Part 1: From Charlemagne to ing.grid

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4.0

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Software availability:

-

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Abstract. Comics journalism constitutes a graphic form of literary journalism. According to John S. Bak and Christopher Craig [1], comics journalism and literary journalism share key features: they both read like fiction but are based on immersive research, eyewitness accounts, and verifiable facts scrutinized as closely as possible. RWTH Aachen University has recently embraced comics journalism as a means of communicating ongoing research at the Chair of AI Methodology within the Faculty of Computer Science [2]. To enable close observation and immersive documentation of scientific practice, comic artist Alfred Neuwald was invited to work alongside the researchers. The resulting collaboration established the general framework for this project. Here we present the first result of a modified artist-in-residence concept implemented by the German National Research Data Infrastructure for Engineering Sciences (NFDI4ING).

1 Introduction

Inviting an outsider to observe and take part in scientific practice has a methodological history that can be traced back to the ethnographic approach of Bruno Latour and Steve Woolgar's *Laboratory Life* [3], [4]. *Laboratory Life* is a foundational work in science and technology studies. The book includes a floor plan of the research laboratories, offices, and meeting rooms in which Latour worked alongside the researchers. It also reproduces photographs of the working environment and, in some cases, of the researchers. Latour and Woolgar thus document interactions with and among the researchers in a spatially situated manner. They report on the verbal content of these interactions and the vocabulary used. Their main focus is the researchers' investment in the iterative production process that brings about the principal outcome of scientific work: peer-reviewed publications. Crucially, Latour and Woolgar also provide transcripts and occasionally accompany them with in-line commentary. The in-line commentary is used to describe aspects of spatially situated interaction that are not captured in the transcripts themselves, including deictic gestures—gestures that establish spatial reference within a shared environment. A key constraint of in-line commentary, however, is that it competes for space with the transcript itself: as descriptions of embodied and spatially situated action become more detailed, the transcript becomes increasingly fragmented and more difficult to read. To address this problem,

18 multimodal forms of transcription typically move such commentary into a separate column at
19 the page margins (see, for example, [5]). This layout allows verbal interaction and descriptions
20 of embodied action to be presented simultaneously without interrupting the flow of the transcript.
21 Moreover, this layout permits the integration of graphical representations alongside the transcript,
22 which constitutes a comparatively recent development in conversation analytic research (see, for
23 example, [6]).

24 Comics represent embodied speech. They do so by depicting speakers and listeners within
25 their environment together with spoken utterances, deictic gestures, and, in some cases, internal
26 thoughts represented through thought balloons. As a graphic medium, comics are therefore
27 particularly well suited to documenting multimodal interaction in situated research environments.
28 Comics, however, usually transport readers into the realm of imagination rather than documented
29 fact. Comics journalism—together with its European and Japanese counterparts, *BD reportage*
30 and documentary manga—differs precisely in this respect [1], [7].

31 At RWTH Aachen University, comics journalism has recently been embraced as a means of
32 communicating ongoing research at the Chair of AI Methodology within the Faculty of Computer
33 Science [2]. The collaboration between researchers and comic artist Alfred Neuwald established
34 the general framework for the present project. By combining ethnographic observation with the
35 multimodal representational capacities of comics, we explore comics journalism as a method for
36 documenting situated research practices within the administrative office of the German National
37 Research Data Infrastructure for Engineering Sciences (NFDI4ING).

38 2 Result

39 2.1 A FAIR Comic on Research Data Infrastructure

40 **Dear Reviewers, please note:** The line numbers in the following pages refer to comic panels,
not to individual text lines.

From Charlemagne to Little Charles.
 From Carolingian minuscule via Middle Francia to  NFDI4ING.* Research data
 infrastructure is and remains an exciting subject.



* NFDI4ING is the consortium for engineering sciences within the German National Research Data Infrastructure (NFDI)

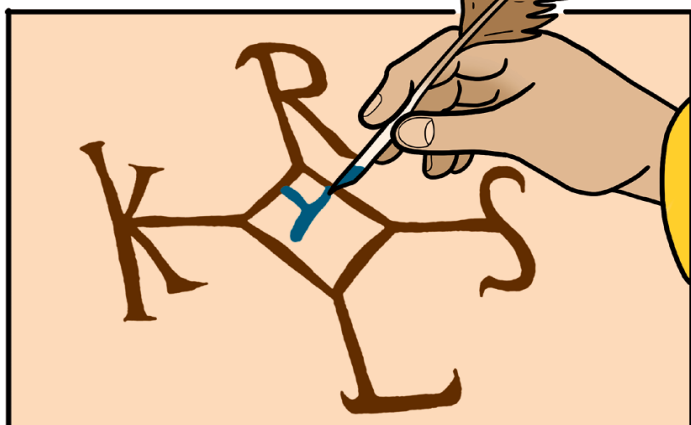
43

In the 8th/9th century, Charlemagne began building a sustainable knowledge infrastructure through his centralized culture of writing. He relied on a network of scribes in monasteries and at his court in Aachen.



44

Charlemagne could not write. When signing documents, he merely added the so-called completion stroke to the Karolus monogram.



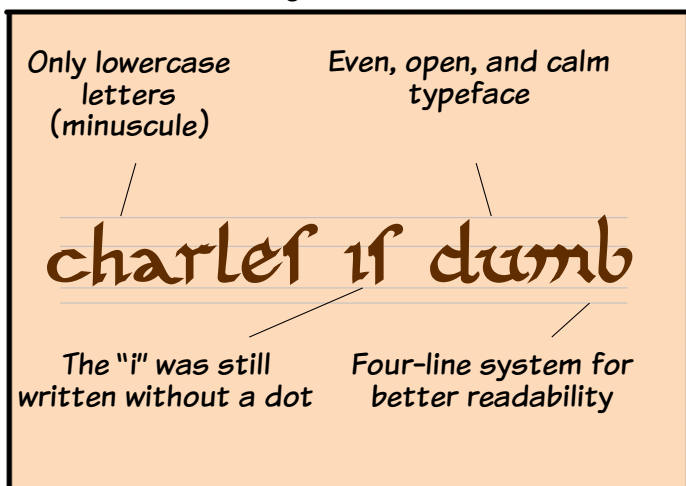
However, Charlemagne promoted education by having scholars dictate, copy, and distribute documents.

45

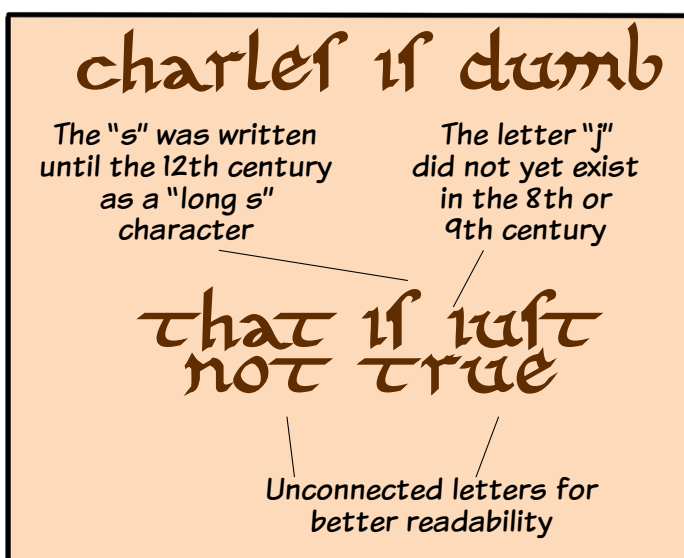


46

To ensure the readability and consistency of documents throughout his vast empire, he commissioned the development of the "Carolingian minuscule".

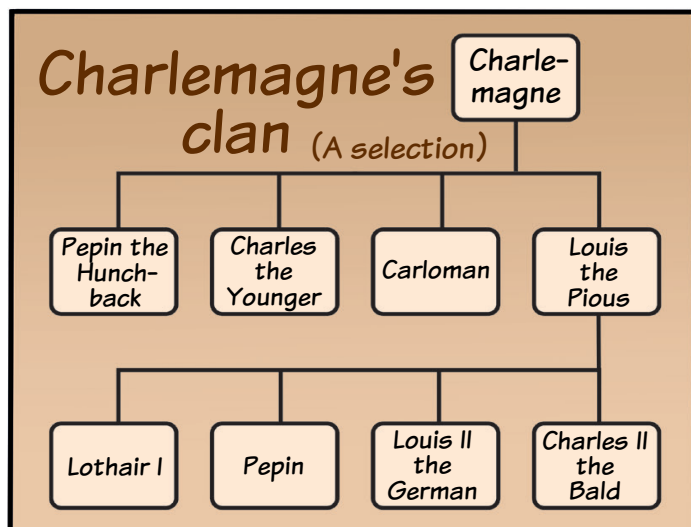


47



48

Charlemagne's sons and grandsons continued this analogue research infrastructure based on parchment and ink.



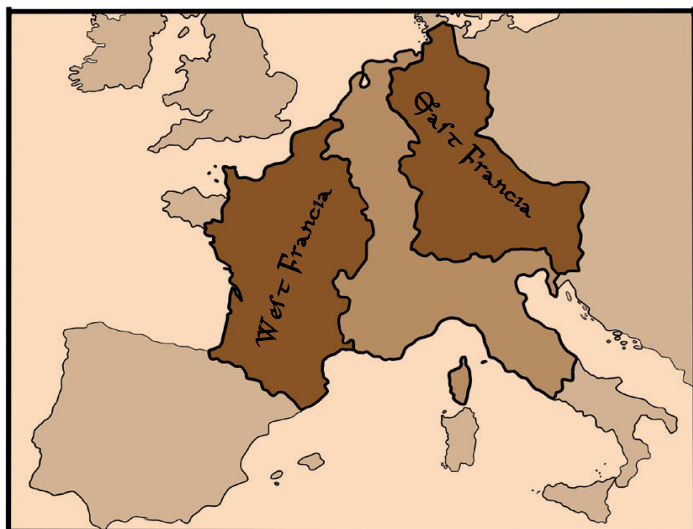
49

One of his grandsons, Lothair I, played a major role in the Treaty of Verdun in 843.



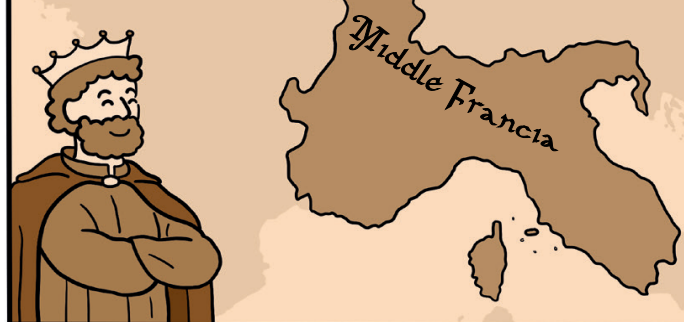
50

This treaty is regarded as the birth of the predecessors of France and Germany (West Francia and East Francia).



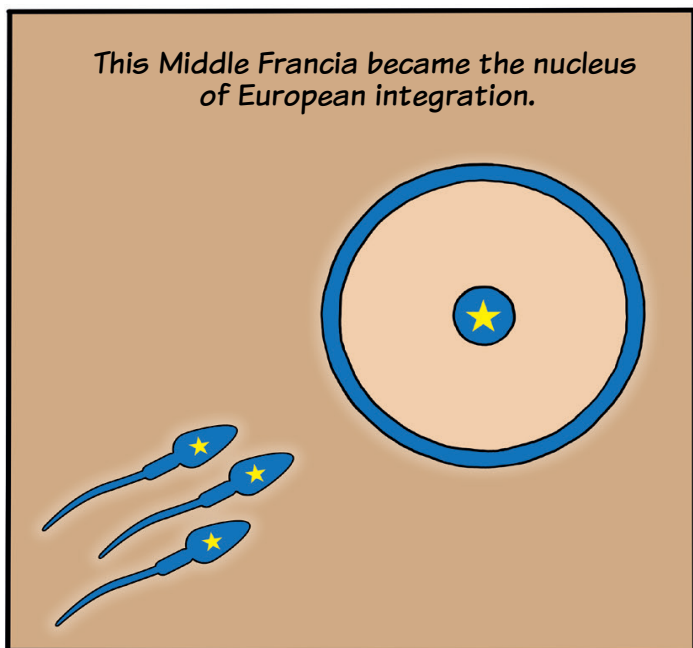
51

Lothair himself received Middle Francia. It stretched as a long, narrow corridor from the North Sea to the Mediterranean.



52

This Middle Francia became the nucleus of European integration.



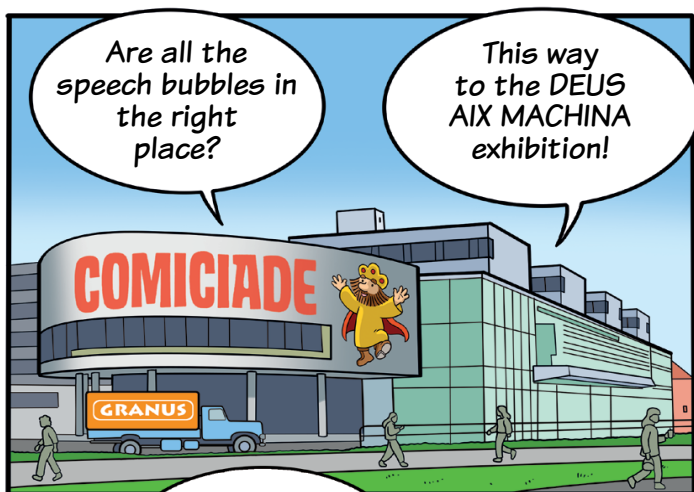
53

Many institutions of the European Union are now located along this former corridor (Brussels, Luxembourg, Strasbourg, Rome).



54

And so the Comiciade at the Technology Centre on Europaplatz in Aachen opens its doors to comic fans from all over the world.

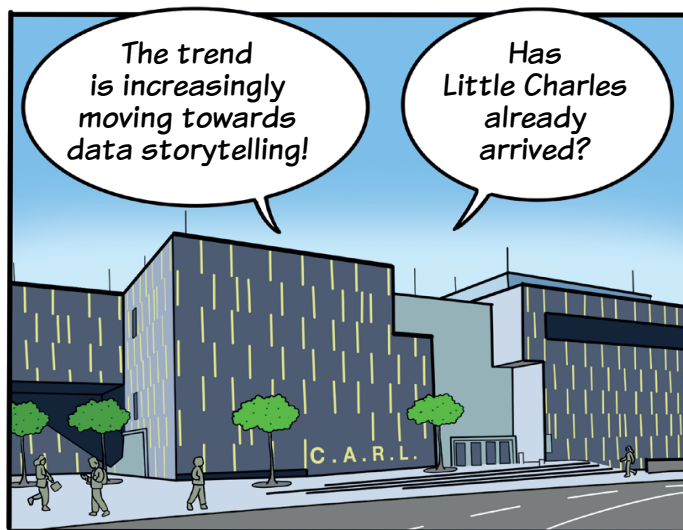


56



Preparations are already under way at RWTH Aachen's C.A.R.L. as well.

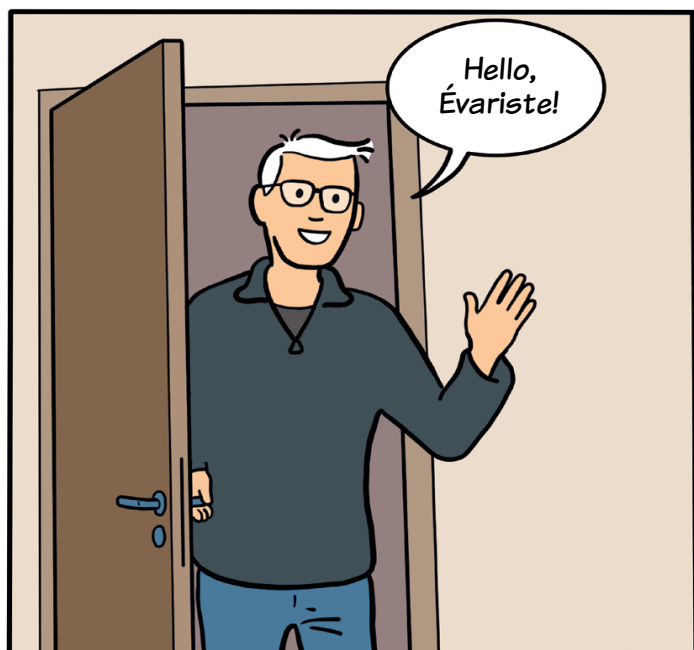
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58



59



60



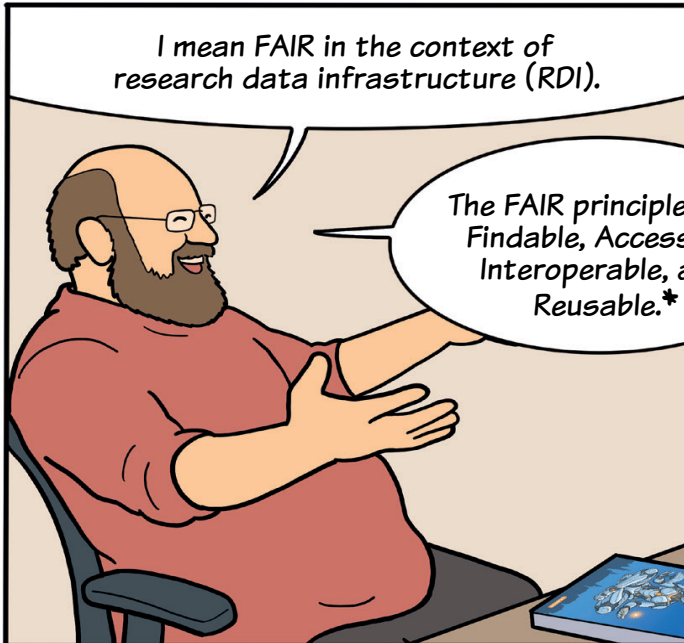
I really enjoyed your comic about your time at the Chair of Methodology of Artificial Intelligence. What do you think about making the world's first FAIR comic now?

61



The world's first FAIR comic ever?

62



I mean FAIR in the context of research data infrastructure (RDI).

The FAIR principles are: Findable, Accessible, Interoperable, and Reusable.*

63



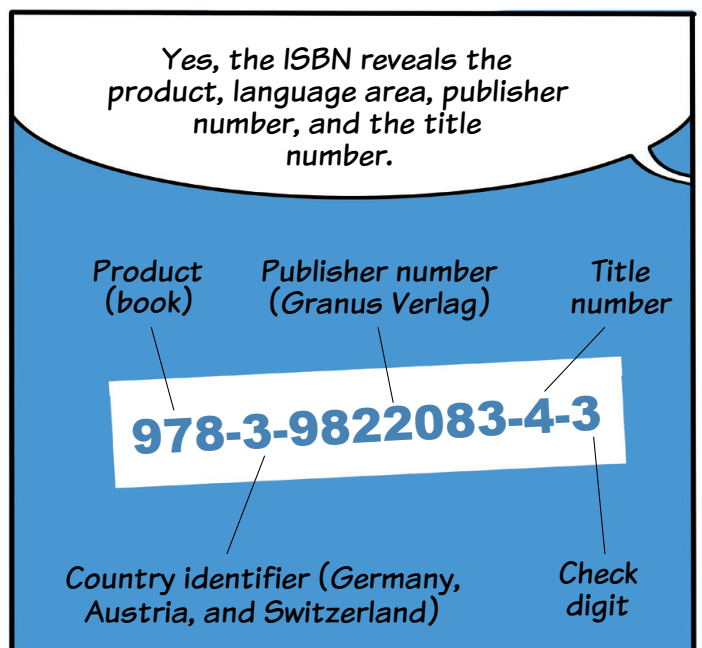
You'll have to explain that to me, Évariste.

64



It's quite simple. On the back of your comic, you included an ISBN so it can be found more easily in bookshops.

65



Yes, the ISBN reveals the product, language area, publisher number, and the title number.

Product (book)

Publisher number (Granus Verlag)

Title number

978-3-9822083-4-3

Country identifier (Germany, Austria, and Switzerland)

Check digit

*Findable, accessible, interoperable, and reusable

66

You need an ISBN to be included in the German Books in Print directory (VLB). If you are not listed there, it is very difficult to get your book into shops.



67

The VLB is the central, independent database and research tool for the German-language book trade.



68

Interesting. We researchers use a DOI (Digital Object Identifier) instead of an ISBN.



69

It is a unique, persistent identifier for objects of all kinds. An object may have several DOIs, but a DOI always points to only one object.

DOI resolver

Repository

<https://doi.org/10.5281/zenodo.19468332>

Prefix

Or as a QR code



Object

70

A partner of NFDI4ING, the DataCite association, assigns DOIs.*



71

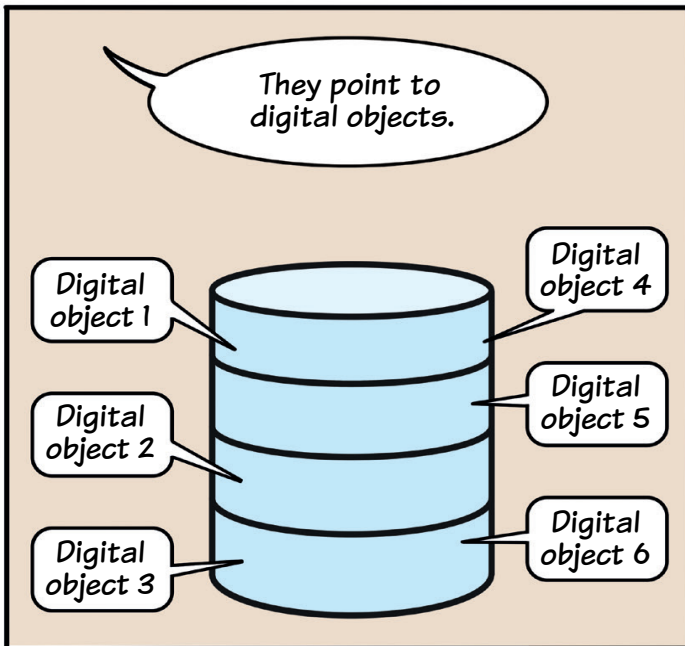
DOIs are the standard in academic publications for precise source references.

And how exactly do they work?



*DataCite is a registration agency for DOIs

72



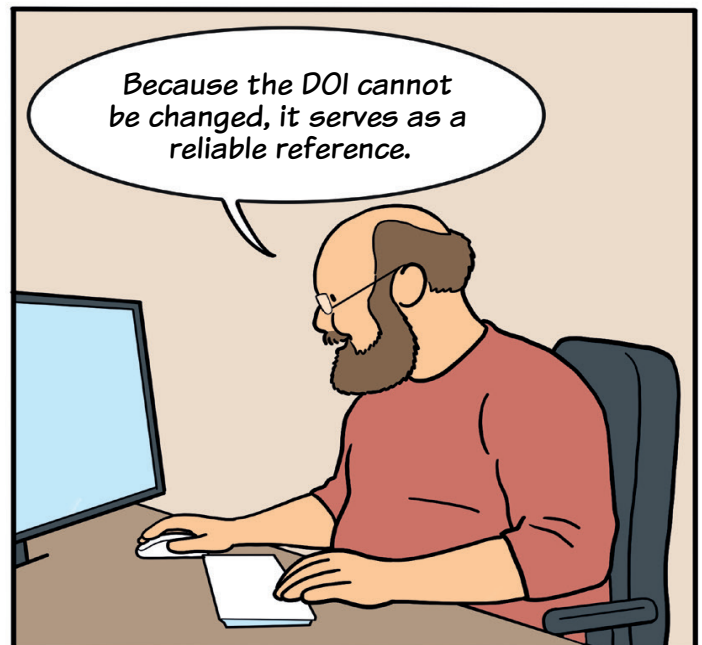
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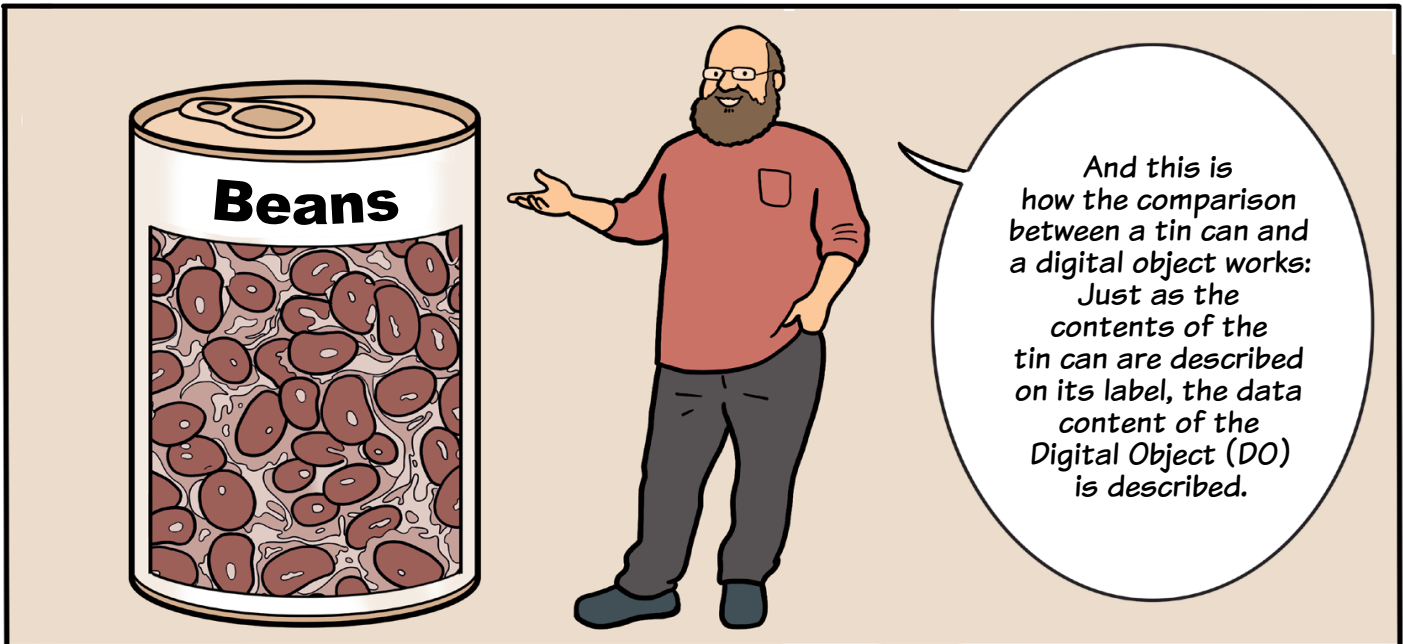
76



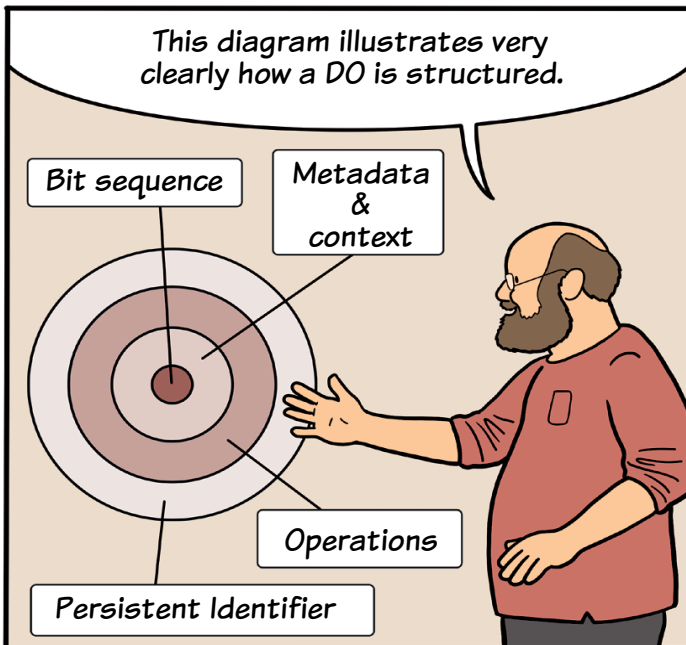
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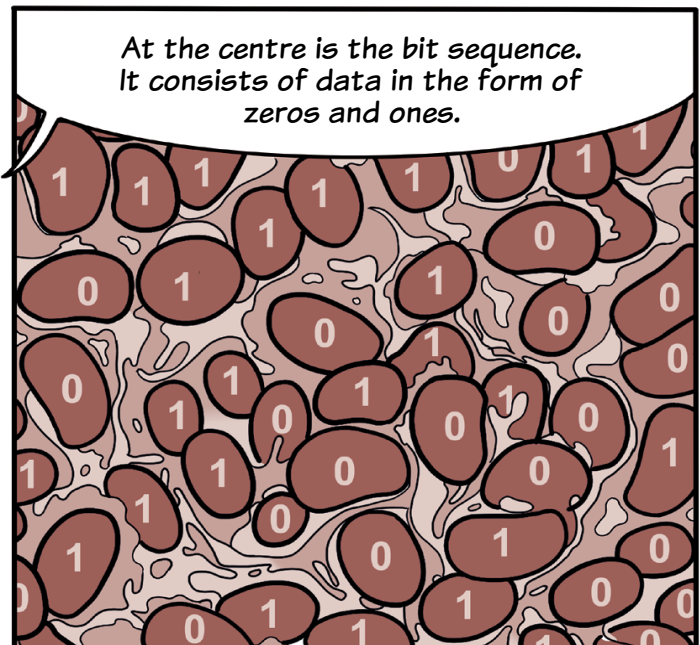
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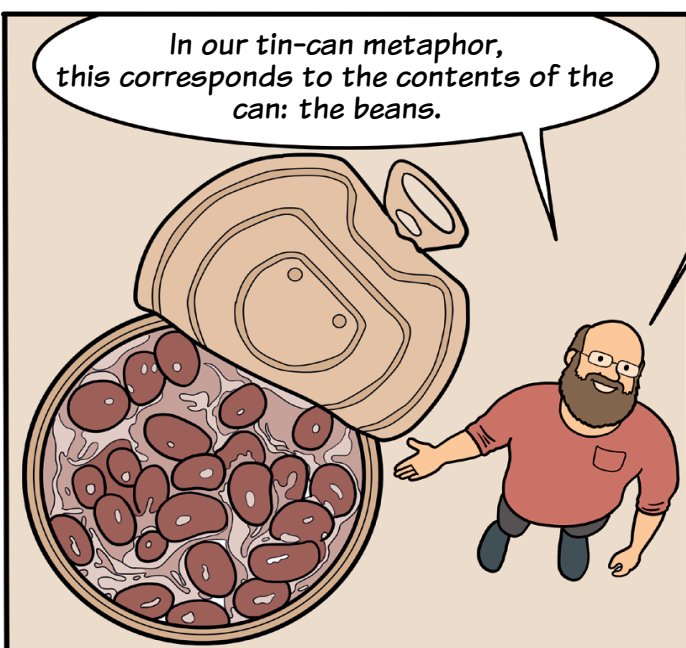
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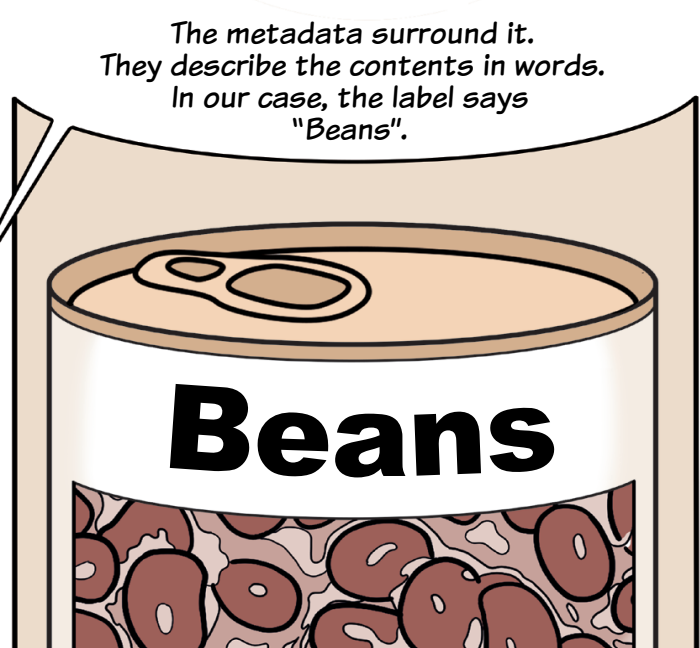
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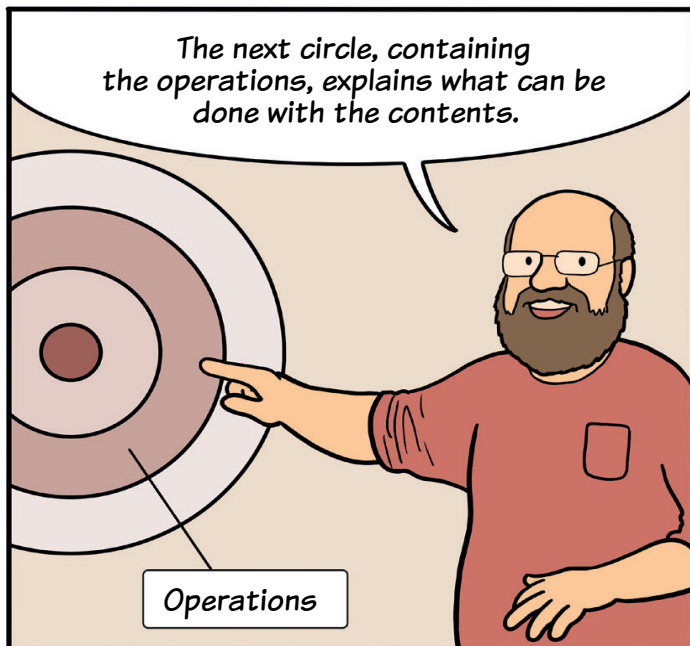
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82

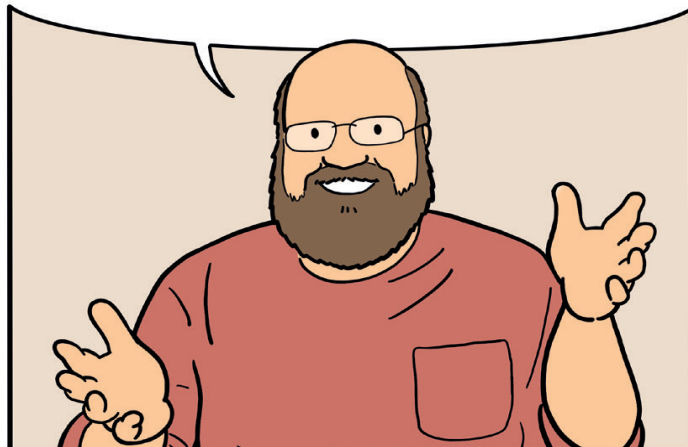


83

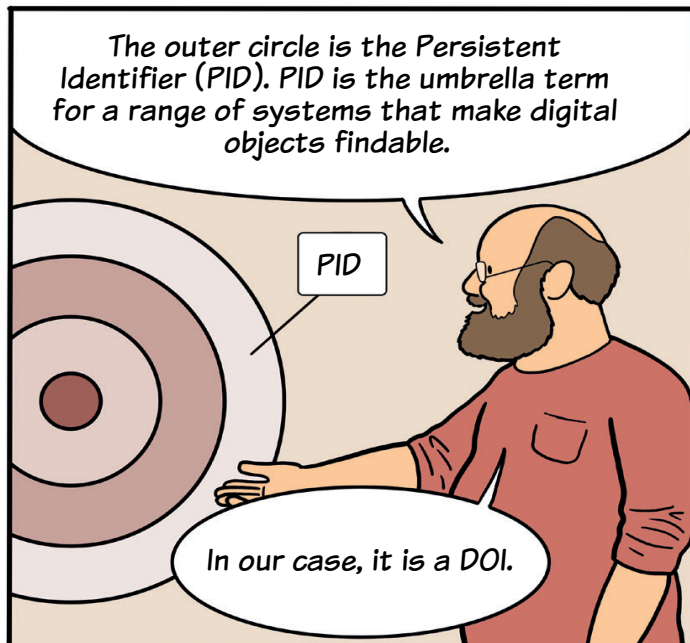


NFDI4ING is working to define more than just a download function within the operations.

84



85

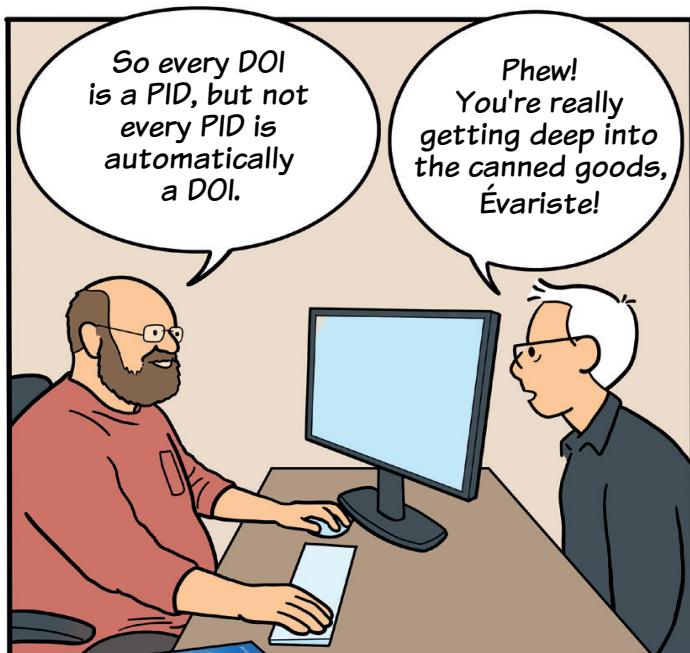


However, there are also other systems, such as URN, Handle, or ORCID.*

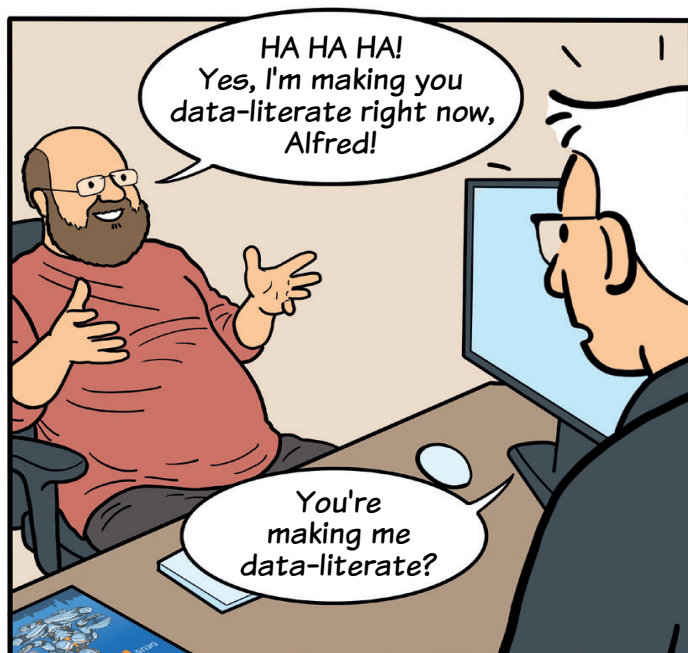
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87



88



*URN = Uniform Resource Name, Handle = Handle System; ORCID = Open Researcher and Contributor ID

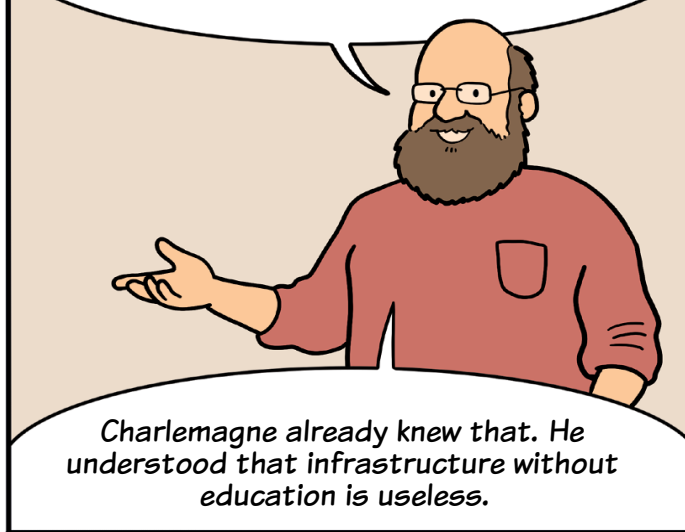
89

In RDI, the term "data literacy" describes the process of acquiring the fundamental skills and knowledge required to read and write research data.



90

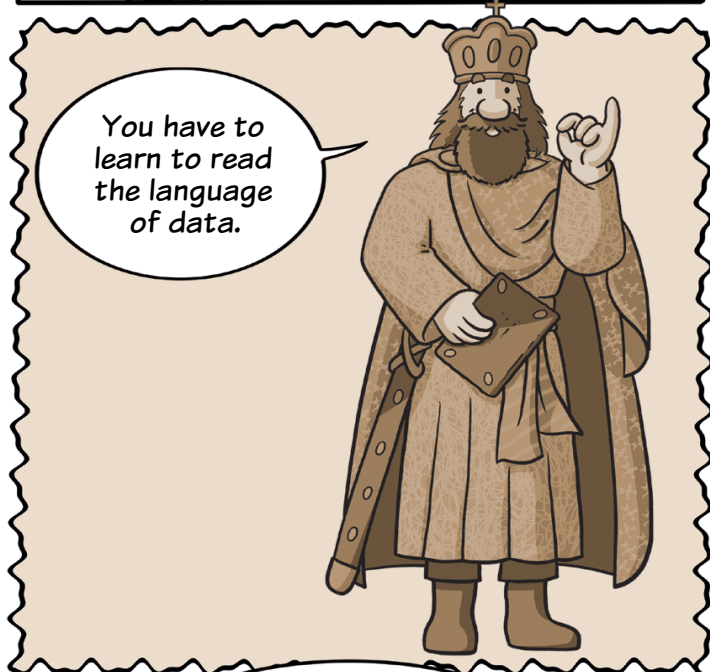
If nobody can read them, even the best archive is useless, and the data remain silent.



Charlemagne already knew that. He understood that infrastructure without education is useless.

91

You have to learn to read the language of data.



92

Charlemagne placed great importance on clerics and administrative officials being able to read and write---also so that they could copy texts correctly. Faulty data transmission was already a problem back then.



93

In other words, copy-and-paste errors in the 9th century.

HA HA HA!
HA HA HA!
HA HA HA!

Now that's what I call successful interoperability between the past and the present!



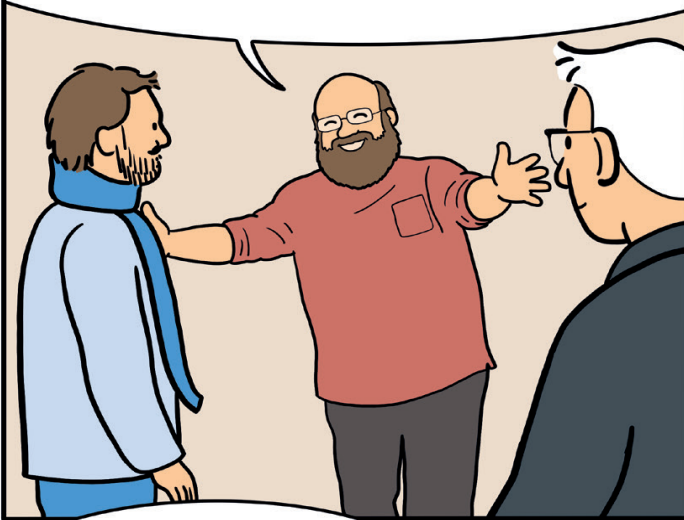
94

What are you laughing about?



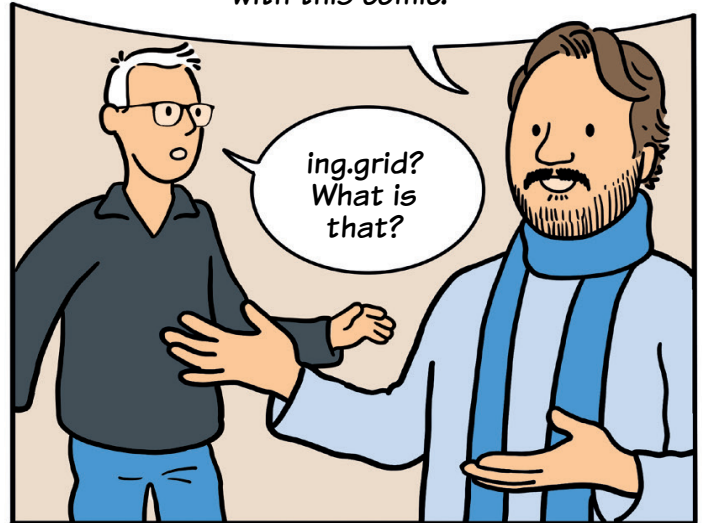
95

Hello, Tobias. We are currently creating a FAIR comic, and it is proving very entertaining.



96

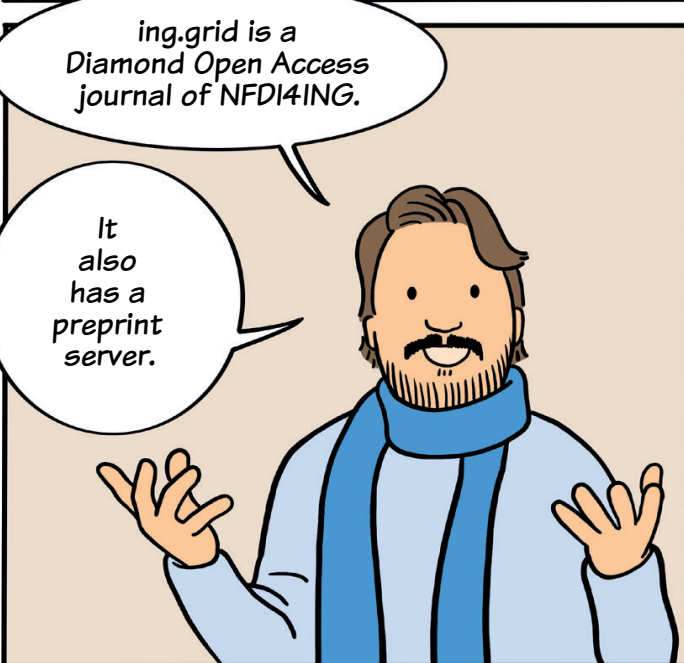
That sounds exciting. Yesterday, I submitted my latest paper to ing.grid. You could do the same with this comic.



97

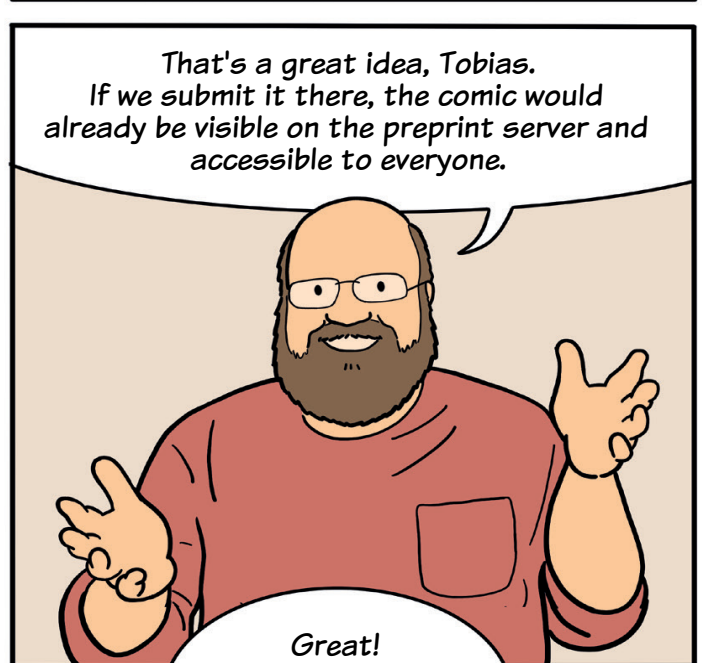
ing.grid is a Diamond Open Access journal of NFDI4ING.

It also has a preprint server.



98

That's a great idea, Tobias. If we submit it there, the comic would already be visible on the preprint server and accessible to everyone.



99

Exactly! A community review would be the first step there. That means the comic could already be commented on and evaluated.



100

Great! Where should I put my completion stroke?



101

Just a moment!
First, we have to translate the comic into
the target language. There is already a
guide for that.



102

This guide can be found
under this HANDLE.

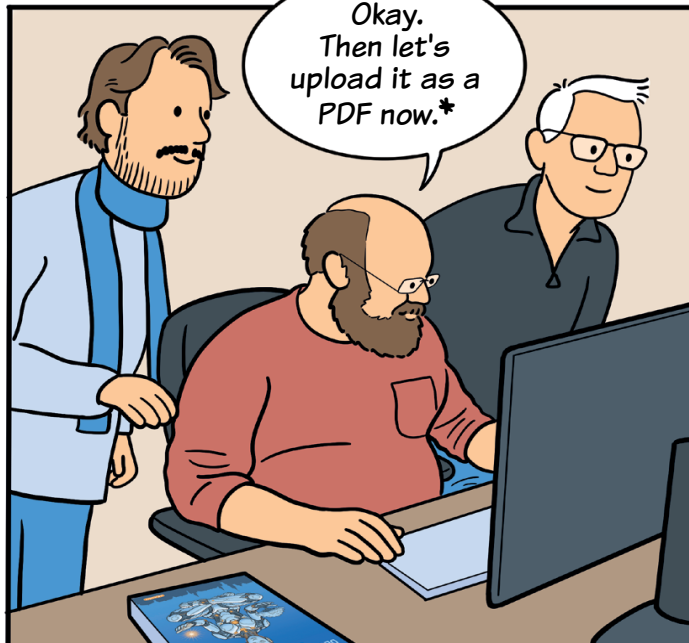
<http://hdl.handle.net/2268.2/12918>

The citation looks like this: [8]



103

Okay.
Then let's
upload it as a
PDF now.*



104

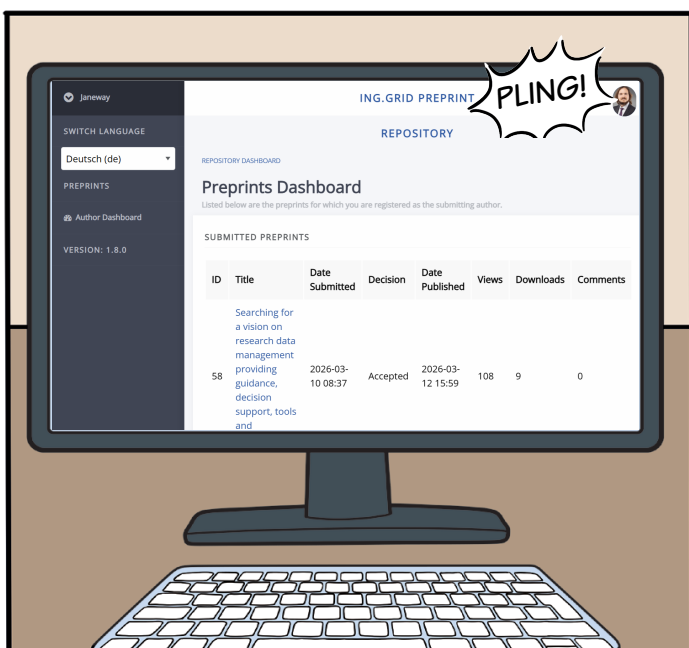
Aaaand...
Here we go!

CLICK



Now it gets
exciting!

105



106

End of
part one



106 3 Materials and Method

107 3.1 Project Development and Collaboration

108 Évariste Demandt first contacted Alfred Neuwald in July 2025 and met with him shortly thereafter.
 109 During this initial meeting, he learned about Alfred's ongoing collaboration with the Chair of AI
 110 Methodology. As this collaboration was expected to continue for several more months, Alfred's
 111 availability for additional commitments was limited. Nevertheless, both parties agreed to remain
 112 in contact and continue exchanging ideas. Évariste suggested that Alfred might participate as a
 113 guest observer in the 2025 Conference on Research Data Infrastructure (CoRDI 2025), and Tobias
 114 Hamann subsequently extended a formal invitation. CoRDI 2025 took place at RWTH Aachen
 115 University from 26 to 28 August 2025 [9]. Alfred participated in the conference, including the
 116 evening event at the Laboratory for Machine Tools and Production Engineering (WZL). He used
 117 the opportunity to familiarize himself with the field of research data infrastructure and with the
 118 people involved.

119 Following the conference, Évariste convened a meeting between Alfred and representatives
 120 of NFDI4ING, IT Center and WZL, as well as a representative of the Chair of Society and
 121 Technology at RWTH Aachen University. It quickly became apparent that we lacked the resources
 122 to replicate Alfred's ongoing collaboration with the Chair of AI Methodology, a project that
 123 enabled Alfred to work full-time with the researchers. As a result, we began exploring alternative
 124 formats for collaboration. Recognizing Alfred's ability to ask questions, Évariste concluded that
 125 a format centered on regular conversations would best accommodate the budget constraints. A
 126 weekly *jour fixe* was therefore scheduled, during which Alfred visited NFDI4ING's administrative
 127 office at the IT Center.

128 Alfred began work on the FAIR comic in February 2026. He photographed Évariste and, later,
 129 Tobias as reference material for the comic, which he created outside RWTH Aachen University
 130 at his atelier. The collaboration followed a meeting-only format: no emails, phone calls, or
 131 messages were exchanged between meetings in order to keep the workload manageable. This
 132 meant that Alfred always came in well prepared with questions and, later, drafts. Meetings were
 133 open-ended and typically exceeded one hour. Alfred's questions sometimes revealed gaps in
 134 shared understanding of research data management (RDM) or exposed issues that could not be
 135 resolved immediately. In such instances, the questions were adjourned and additional expertise
 136 was sought from specialists. This occurred twice during the project: once with support from
 137 NFDI4ING's Task Area Federated Interactive Data Space and once with support from the Chief
 138 Technical Officer of the IT Center.

139 During the weekly meetings, learning was reciprocal. Alfred shared his extensive knowledge of
 140 comic culture, referring to documentary works such as Art Spiegelman's *Maus* [10], Marjane
 141 Satrapi's *Persepolis* [11], and Guy Delisle's *Pyongyang* [12]. Delisle's work, in particular,
 142 received special attention as an outstanding example of observer commentary. Alfred admired
 143 Delisle's ability to document a social world and occasionally drew parallels between his own
 144 encounters with researchers and Delisle's experiences working in North Korea.

145 3.2 Documentary Basis and Narrative Framing

146 This is a work of literary journalism, which means that it is grounded in Alfred's research. The
 147 FAIR comic conveys his first-hand observations, conversations, and photographic documentation
 148 gathered during the weekly meetings at the administrative office of NFDI4ING at the IT Center
 149 of RWTH Aachen University. Among the materials that informed Alfred's work was a leaflet
 150 published by the Landesinitiative NFDI der Digitalen Hochschule NRW and the AG FDM
 151 Awareness [13]. Copies were available in the administrative office, and the leaflet became a
 152 recurring point of reference in conversations between Évariste and Alfred and, later, between
 153 Évariste, Tobias, and Alfred. The influence on the FAIR comic can be seen both in the range
 154 of subjects covered and in the canned-food metaphor, which originates on page 12 of the
 155 leaflet. Another material used by Alfred was a slide desk from NFDI4ING's Task Area Federated
 156 Interactive Data Space with a schematic representation of the different layers that constitute a
 157 FAIR Digital Object.

158 A point of contention was whether fictional elements should be included at all. At the outset of
 159 the project, Évariste was unfamiliar with the distinction between fictional comics and comics
 160 journalism and proposed narrative devices drawn from fictional comics. Alfred generally resisted
 161 such proposals. The need to reconcile them with the request to begin the story in 843 AD
 162 ultimately led to the inclusion of a few fictional elements on the first three pages, which are
 163 examined below. Before turning to these elements, however, we first explain Évariste's request
 164 to begin the story in 843 AD.

165 At the time of writing, Aachen has presented the International Charlemagne Prize for seventy-
 166 six years. First awarded on 14 March 1950, the prize honours individuals who have made
 167 outstanding contributions to peace, unity, and cooperation in Europe [14]. Its creation coincided
 168 with a formative decade of European integration. The decade witnessed the establishment of
 169 the European Coal and Steel Community (ECSC) through the Treaty of Paris (1951; in force
 170 from 1952) and, later, of the European Economic Community (EEC) and the European Atomic
 171 Energy Community (EURATOM) through the Treaties of Rome (1957; in force from 1958). Less
 172 frequently noted is that the combined territory of the six states that founded the ECSC, the EEC,
 173 and EURATOM overlaps substantially with the core regions of the Carolingian Empire. This
 174 overlap is noteworthy because Aachen is closely associated with Charlemagne, who established
 175 a palace complex (Pfalz) there and made the city a center of Carolingian rule. Aachen therefore
 176 occupies a distinctive place in both the symbolic landscape of post-war European integration
 177 and the historical geography of the Carolingian world.

178 The Carolingian connection is also visible at RWTH Aachen University. It is reflected, among
 179 other things, in the acronym C.A.R.L., the name of a representative university building that
 180 served as the venue for CoRDI 2025 [9]. Moreover, the local host of CoRDI 2025, Robert
 181 Schmitt of NFDI4ING, referred explicitly to the Carolingians in his opening remarks of the
 182 CoRDI. Against this backdrop of Carolingian references, Évariste was reminded of a familiar
 183 narrative convention in comics: the use of an introductory map, as found, for example, on the
 184 front endpapers of *Astérix* albums [15]. This convention inspired the idea of opening the FAIR
 185 comic with a map of the Carolingian Empire.

186 The first three pages of the FAIR comic are dedicated to establishing context and continuity

(cf. chapter five in [5]) within the broader geographical and cultural landscape of NFDI4ING. They are also intended to entertain readers and draw them toward the more technical aspects of research data infrastructure. To this end, a small number of fictional elements were introduced, including reconstructions of what Charlemagne and his descendants might have said and done up to the partition of the empire under the Treaty of Verdun in 843 AD. In designing and drawing the first three pages, Alfred used expertise he had acquired in the course of an earlier project [16]. The map of the Carolingian Empire, specifically, is based on a historical map distributed through Alamy [17]. On a more technical note, most of the details of the historical map were omitted in order to emphasize selected features, while a few others were added. Among these is the empire's eastern frontier, which is marked on the map by the added detail *limes sorabicus*. The territories beyond this frontier were subject to varying degrees of Carolingian rule and correspond broadly to the toponomastic boundary between regions characterized by Germanic and Slavic place-name strata [18]. Another addition is the city of Darmstadt, home to Technische Universität Darmstadt, applicant institution of NFDI4ING during its second funding period [19].

As soon as the FAIR comic shifts from the Carolingian past to present-day Aachen and C.A.R.L., the staged setting of the meeting between Alfred and Évariste, the colour palette switches to indicate a different register: conversation. The ensuing dialogue depicts discussions that took place over the course of almost five months of weekly meetings involving Alfred and either Évariste alone or Évariste and Tobias together. The FAIR comic organizes these discussions thematically and condenses them into nine pages.

3.3 Manuscript Preparation and Translation

The process of outlining, designing and refining the comic was conducted in German. As such, the first draft was written in German. As we approached the publication of the comic within ing.grid, the comic had to be translated into English language. However, to make the comic more accessible, we considered different language versions as quite relevant. One possibility of achieving this multilingual solution, Alfred would have to redraw all texts within the comic in a new language. This would have led to long review loops as well as a big overhead for Alfred.

Tobias instead came up with a different, more automated version. He asked Alfred to create a blank PDF version of the comic, containing only the persons, maps etc. but no text. With this blank version, a LaTeX document could be created. In this document, special pages were declared. These pages form full page TikZ¹ images where the blank comic pages form the background of the page. Then, the original German text written by Alfred was placed on top of the blank background PDF as TikZ nodes. These nodes can be styled in different ways, allowing for different fonts and font sizes as well as special text forms e.g. curved or rotated texts. The nodes can be freely placed on the page and allow for precise text placement of the comic text. Furthermore, manual line breaks can be placed in the language file. In that way, the text can manually be adjusted to enhance styling and balance of the text as seen fit by the artist.

Yet, another problem arose. The image of three-dimensional toy bricks forming the letters “FDI” for the German “Forschungsdateninfrastruktur” on the first page also had to be translated. For

1. TikZ is a recursive acronym for “TikZ ist kein Zeichenprogramm”, which is German for “TikZ is not a drawing program”.

this, different language versions of the image were generated by artificial intelligence, namely OpenAI’s ChatGPT 5.5 version. The AI was asked to exactly keep the style of the image but change the original hand-sketched “FDI” to “RDI”. The original “FDI” was also removed from the blank comic version so that the background would not interfere with the overlay image that is language dependant.

In a similar manner, the language files can automatically be translated using Artificial Intelligence (AI). To proof this concept, we let AI translate the comic into latin and elvish. In that manner, all dextrograde languages, i.e. languages that are read from left to right, can be displayed in the comic. Elvish furthermore proves, that languages with non-latin letters can be displayed as well. Sinistrograde (right to left writing) or boustrophedon (linewise changing of writing direction) writing was not yet implemented nor tested.

The source code can be found on [GitHub](#). As a wrapper, the file `pages_functions.tex` is processing both the different `language_files/XY.tex` and their corresponding `pages_XY` files. Every `language_files/XY.tex` contains all text lines ordered by their appearance on the pages in the corresponding language XY, which allows for a translation of texts independently from their appearance in the document. Also, manual line breaks can be added here as needed. The `pages_XY` files handle the positioning and scaling of texts for each language XY individually, which accounts for different lengths of the same contents in different languages. The language is chosen in the `main.tex`, e.g. with the command `\newcommand{\FDILanguage}{en}`, an English version will be generated. As of writing, the original German version and this translated English version exist.

4 Discussion

4.1 Dialogue as a Documentary and Didactic Form

The conversational format of the FAIR comic has a philosophical pedigree reaching back to the dialogues of Plato (*circa* 424/3–348/7 BC [20]). Parallels include the use of historical interlocutors [20] and a characteristic division of labour: Socrates asks questions, while his interlocutors formulate and defend answers. Like Socrates, Alfred frequently advances the discussion through questions. One of the most arresting things about the Platonic dialogues is that most of them have survived for more than two millennia and continue to be widely read and discussed today. We contend that such an unusually long period of transmission should be taken as an indication of the longevity not only of their content but also of their form. References to Plato continue to appear in contemporary culture and education. Examples include Jay-Z, Kanye West, Frank Ocean, and The-Dream’s song “No Church in the Wild”, which invokes the *Euthyphron* dilemma [21], [22]; Helme Heine’s children’s book *Tante Nudel, Onkel Ruhe und Herr Schlau*, which adapts the *Politeia* [23]; Höfner and Süßbier’s didactic comic *Das verrückte Mathe-Comic-Buch*, which includes a re-telling of the *Menon* [24]; as well as James Orr’s lecture series *Plato* [25]. This continued presence suggests not only the enduring relevance of the subject matter, but also the effectiveness and remarkable adaptability of the dialogical format through which that subject matter is explored. As Orr points out in the opening lecture of his series, “once you know all of the sides, you understand your own” [25, at 55:09]. The FAIR

comic similarly relies on dialogue as a means of exploring and communicating complex, albeit technological, matters. We hope that readers interested in research data infrastructure will be profitably entertained by the result.

We have introduced the FAIR comic by reference to the documentary practices of ethnographic research [3], [4], and discourse and conversation analysis [5], [6]. More specifically, we have situated comic culture within a continuum of practices concerned with documenting deixis and embodied speech (see 1). We have argued that, as a graphic medium, comics are particularly well suited to documenting multimodal interaction in situated research environments. In this sense, the FAIR comic may itself be understood as a documentary form of speech. If we have now reached back to the dialogues of Plato, it is still in an attempt to reflect on the structure of the FAIR comic and, more generally, on what helps readers understand the spoken word [26].

4.2 The FAIR Comic between Research and Science Comics

We conclude our discussion by comparing the FAIR comic to related forms of comic culture, such as comics about researchers in research institutions (see, for example, Jorge Cham's *PHD Comics* or Eppendorf's *Lab Life*; [27], [28]), as well as science communication comics that explain fields of research (see, for example, Larry Gonick's *Cartoon Guide* series or Matteo Farinella's *Neurocomic*; [29], [30], [31]). The FAIR comic sits comfortably between these two broader lineages and follows a journalistic ethos. What we seek to achieve is not adequately captured by saying that the FAIR comic is about research practice, nor even that it is researched, but rather that it is explicit about the process of learning—or coming to understand. To put our claim in direct speech: "This is what we discussed while trying to establish a mutual understanding of research data infrastructure."

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6 Roles and contributions

Alfred Neuwald: Conceptualization, Methodology, Writing, Visualisation

Tobias Hamann: Conceptualization, Methodology, Writing, Software

Évariste Demandt: Conceptualization, Writing

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